



Nutrition Interventions for Under-5 & Women of Reproductive Age

LLM integrated large scale rapid evidence review process

1,996

Papers Ranked

200

Reviewed (117 full text)

15

Interventions Shortlisted

3

Evidence Tiers

PubMed + OpenAlex systematic search · population-targeted (under-5 / WRA) · Pipeline v3.0 · June 2026

Why a Pipeline — Rapid Review at Scale

A human rapid review is time-intensive

- ✗ Screening **thousands of abstracts by hand** — a single-question rapid review typically takes weeks to months
- ✗ Manual extraction of effect sizes, confidence intervals and study designs, paper by paper
- ✗ Synthesising across **15 interventions x hundreds of papers** compounds the workload
- ✗ Screening and scoring choices are hard to document, audit and reproduce

... but an unconstrained LLM is unreliable

- ✗ **Hallucinates** effect sizes and fabricates citations
- ✗ **Leaks figures from training data** with no provenance
- ✗ Cannot reliably **search, deduplicate or rank** at corpus scale
- ✗ Context limits — it cannot hold thousands of papers at once
- ✗ Misclassifies study types when it pattern-matches on titles

The gap to close

Speed needs automation; trust needs constraints. The pipeline supplies both — deterministic code does the search and ranking, and the LLM is fenced into a verified, citable corpus.

Our Approach — Code Ranks, the LLM Only Reads

Search and ranking are fully automated and **LLM-free**. The LLM enters only at the end, and only over the curated, ranked dataset — never the open web or its own memory.

The process

- ▶ **Automated search** — code scrapes PubMed + OpenAlex with curated MeSH / publication-type queries; **no LLM**
- ▶ **Score + rank** — deduplicate, enrich citations, and score on MeSH + publication type + population; keep the top 200; fully **deterministic**
- ▶ **LLM reads** — the model is handed *only the ranked papers* and pulls information from them, nothing else
- ▶ **Grounded synthesis** — a corpus PMID on every number, then a verifier lints each claim against the corpus

Guardrails

- ✓ The LLM never decides relevance — *what to read* and *in what order* are solved deterministically before it is invoked
- ✓ Its purview is fenced to the ranked dataset — no open web, no training-data figures
- ✓ Prompts enforce systematic-review discipline: study type stated **verbatim** from metadata, a PMID for every figure, and **no cost-effectiveness rating without a real CEA**

Net effect

The speed of automation with the rigour of a systematic review — every claim traceable to a ranked, citable source.

Evidence Prioritization Framework

Figure 1. Fifteen shortlisted interventions positioned by evidence strength (x-axis) against implementation readiness — cost-effectiveness + proven scalability (y-axis). Dashed lines delineate four action quadrants; dot colour indicates evidence tier.



Reading the chart

Strong evidence + high readiness (top-right): strong evidence, a genuine favourable CEA, and a proven delivery platform — six interventions.

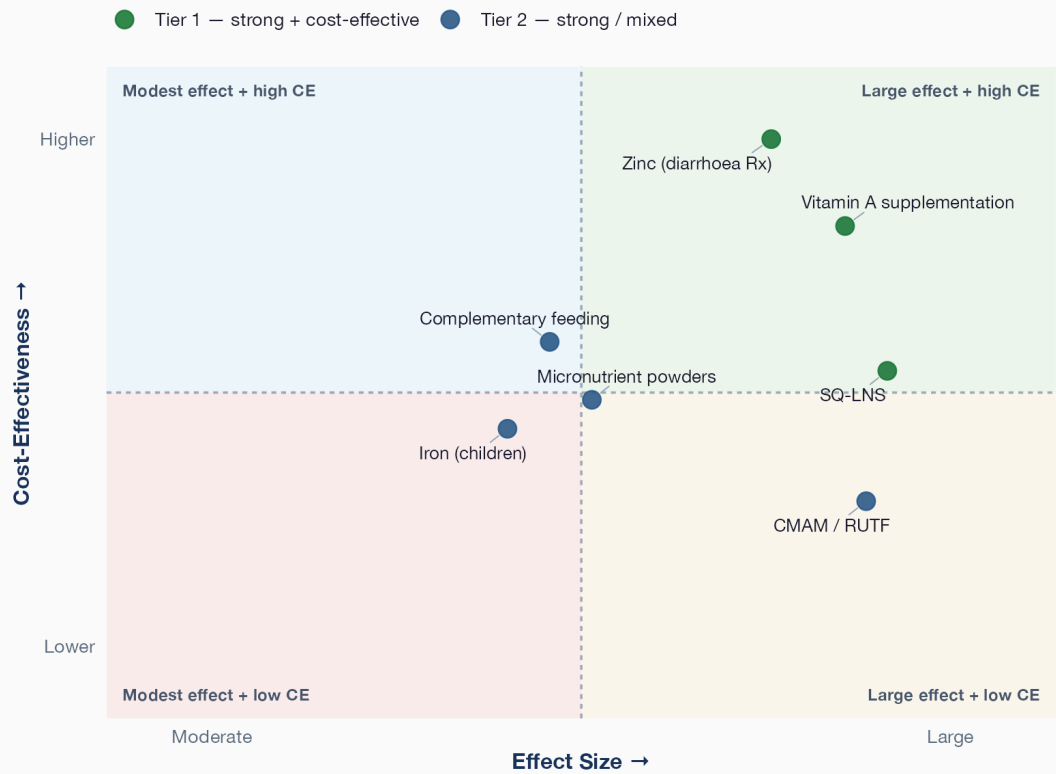
Strong evidence + low readiness (bottom-right): robust evidence but higher unit cost or coverage gaps (CMAM, SQ-LNS, complementary feeding).

Weak evidence + low readiness (bottom-left): WASH and cash transfers — small or indirect nutrition effects, and **no intervention-specific CEA** in the corpus → cost-effectiveness *Unknown*.

Axes are a qualitative synthesis (evidence strength against cost-effectiveness + proven scalability), not a fitted model.

Children Under 5 — Intervention Prioritization

Figure 2. Child interventions (6–59 months) by effect size on the primary outcome (x-axis) against cost-effectiveness (y-axis, lower cost per DALY = higher). Dot colour indicates evidence tier. All effect sizes and costs are corpus-grounded (PMIDs in table).



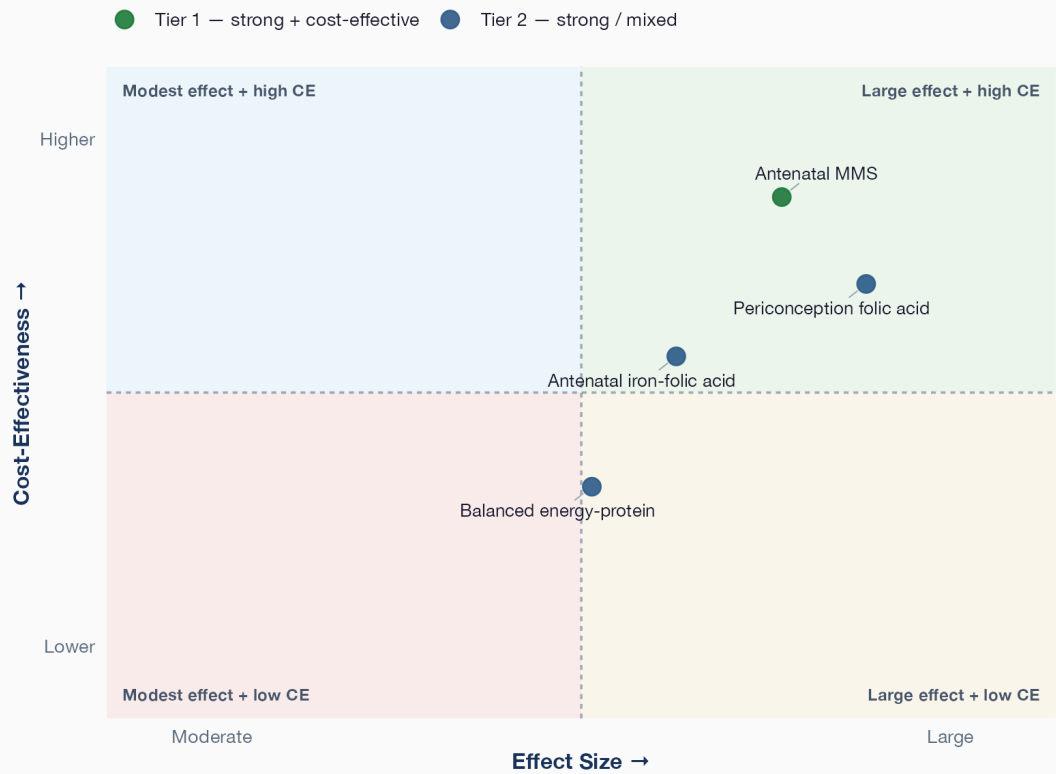
Supporting Data (corpus-grounded)

Intervention	Key effect	PMID
Vitamin A (6–59 mo)	RR 0.76 all-cause mort. (0.69–0.83)	21868478
Zinc — diarrhoea Rx	~23% diarrhoea mortality ↓	20348128
SQ-LNS	RR 0.73 mortality (0.59–0.89)	31697329
Comp. feeding	HAZ SMD 0.23 (0.09–0.36)	24564534
CMAM / RUTF	+51% recovery vs standard	24564235
Iron (children)	Hb ↑ in anaemic; risk if replete	17158406

Effect is conditional on baseline deficiency for VAS, iron and preventive zinc — targeting beats universal provision. CEA per intervention from Phase 2.

Women of Reproductive Age — Intervention Prioritization

Figure 3. Maternal interventions by effect size on the primary outcome (x-axis) against cost-effectiveness (y-axis). Dot colour indicates evidence tier. Antenatal MMS is the headline "transition" opportunity (from iron-folic acid).



The MMS transition

Antenatal **MMS** cuts low birth weight vs iron-folic acid (RR 0.86, 95% CI 0.79–0.93; PMID 21673856) at **\$3–15 per DALY averted** (PMID 35192606). WHO conditionally recommended the IFA → MMS transition.

The adherence bottleneck

Pooled iron-folic acid compliance in sub-Saharan Africa is just **39.2%** (PMID 33852614) — coverage, not efficacy, is the binding constraint.

Supporting Data

Intervention	Key effect	PMID
MMS	RR 0.86 LBW (0.79–0.93)	21673856
Folic acid	LBW OR 0.59; NTD –70%	31680411 / 20348114
IFA	Hb ↑; compliance 39.2%	33852614
Bal. energy-protein	birthweight ↑ (undernourished)	25857334

Interventions by Delivery Platform & Evidence Tier

Table 5. The 15 shortlisted interventions by delivery mechanism and evidence tier. Community-based and food-system platforms concentrate the strongest, most cost-effective options.

Delivery Platform	Tier 1 (Strong + cost-effective)	Tier 2 (Strong/mixed)	Tier 3 (Promising/indirect)
Health Facility <i>ANC / Postnatal</i>	Antenatal MMS Zinc for diarrhoea	Antenatal iron-folic acid Periconception folic acid Balanced energy-protein	—
Community-Based <i>CHW Delivery</i>	Vitamin A supplementation Breastfeeding promotion SQ-LNS	Micronutrient powders Complementary feeding CMAM / RUTF Iron (children)	WASH
Food System <i>Fortification</i>	Large-scale fortification (iron, iodine, folic acid, vit A)	—	—
Cash / Social Protection	—	—	Cash transfers

All six Tier-1 interventions ride existing platforms — community health workers, antenatal care, or the food-fortification supply chain — which is why they pair strong evidence with proven scalability. Coverage and adherence, not efficacy, are the binding constraints.

Large-Scale Food Fortification of Staple Foods

Population: children + WRA

Evidence: A

Cost-effectiveness: High

Scalability: Proven national

Evidence (A)

- ▶ LSFF umbrella SR/MA (vitamin A, iodine, iron, folic acid), *AJCN* 2019: anaemia **−34%** (RR 0.66, 0.59–0.74), goitre **−74%** (OR 0.26), NTD **−41%** (OR 0.59) — PMID 30997493
- ▶ Double-fortified salt (iron + iodine), *Advances in Nutrition*: anaemia RR 0.59 (0.46–0.77); IDA RR 0.37 (0.25–0.54) — PMID 29767699
- ▶ Supporting Cochrane reviews: zinc fortification (PMID 27281654), vitamin A fortification (PMID 31074495)
- ▶ Fortified dairy / cereal (older children): weak, non-significant haemoglobin effect — PMID 30673769

Cost-effectiveness (High)

- ▶ SR of economic evaluations across **63 countries** (>200 analyses): cost-effective for reducing malnutrition across most LMIC settings
- ▶ A genuine intervention-specific CEA was retrieved → rating permitted

Scalability (Proven national)

- ▶ Salt iodization and flour fortification are among the **most widely scaled** nutrition interventions globally
- ▶ Delivered through existing food-industry supply chains with regulatory mandates
- ▶ Minimal per-capita cost; no behaviour change required

What Worked Well

The pipeline auto-drafted syntheses for **all 15 interventions**. To test accuracy, a reviewer audited **one** end-to-end — large-scale fortification — checking every claim against the source corpus. It held up on all five.

What the LLM synthesis claimed	Source	Human verification
Anchor = large-scale fortification umbrella review (Keats, <i>AJCN</i> 2019)	PMID 30997493	✓ Correctly the #1-ranked paper of 1,996; classified as <i>AJCN</i> SR/MA — not mislabelled "Cochrane"
Anaemia –34% (RR 0.66); goitre –74% (OR 0.26); NTD –41% (OR 0.59)	30997493	✓ Verbatim in the abstract (all 95% CIs match)
Double-fortified salt: anaemia RR 0.59; IDA RR 0.37	29767699	✓ Verbatim
Fortified dairy / cereal: weak, non-significant effect	30673769	✓ Honest — null result flagged, not inflated
Cost-effectiveness rated High	63-country CEA	✓ Earned — a genuine fortification CEA was retrieved, not invented

Accuracy verdict

Every reported effect size is grounded in its cited paper — the verifier returns **0 not-in-corpus**. Markedly cleaner than the earlier Vitamin A case: correct study types, no misattribution, cost-effectiveness backed by a real CEA.

Why it held up

The hard decisions were made by **deterministic code, not the model**: the LLM only ever saw a ranked set of real papers, the prompt demanded a corpus PMID for every figure, and the verifier flagged anything unsupported. An unconstrained model asked the same question would likely have invented a cost-per-DALY figure or mislabelled the umbrella review as Cochrane — exactly the failures seen in the earlier Vitamin A synthesis.

Limitations

Gaps found for this intervention

- ✗ **Completeness:** no standalone **wheat-flour** or **rice** fortification review retrieved; salt iodization captured only via double-fortified salt
- ✗ **Below the cut:** two relevant iron-fortification reviews (PMID 32318374, 37665781) are in the database but ranked outside the top 200
- ✗ **No trial-overlap detection:** the umbrella review and the component reviews share underlying trials — apparent independence is slightly inflated
- ✗ **Noisy CEA bucket:** the top cost-effectiveness hit was an off-topic micronutrient-powder study; only the #2 was a true fortification CEA
- ✗ **Minor imprecision:** one SR mis-described as "Cochrane-adjacent"; scalability claims not corpus-cited

But none produced a false claim

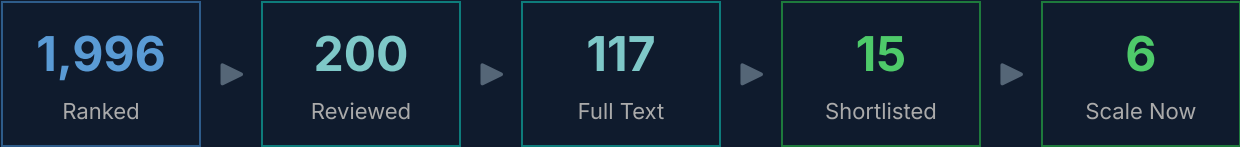
Each issue bounds how *complete* and *self-aware* the synthesis is — none introduced an unsupported number. The grounding rules and the verifier held.

Pipeline-wide caveats

- ▶ Population filter tightens retrieval — some relevant reviews fall below the top-200 cut
- ▶ ~40% of top papers are abstract-only (effect sizes not full-text-verified)
- ▶ **Deferred:** a trial-level evidence graph (shared- / dominant-trial detection) is the next milestone

*Takeaway: the pipeline is accurate on what it reports, but **human review remains essential** for completeness and structural judgement.*

Summary & Key Takeaways



Tier 1 — "Scale Now" Interventions

#	Intervention	Key effect (corpus PMID)	Cost - effectiveness
1	Large-scale food fortification	anaemia RR 0.66 (30997493)	High
2	Vitamin A suppl. (6–59 mo)	RR 0.76 all-cause mort. (21868478)	High (35390077)
3	Zinc for diarrhoea treatment	~23% diarrhoea mort. ↓ (20348128)	High (~\$12/DALY)
4	Antenatal MMS	LBW RR 0.86 (21673856)	High (\$3–15/DALY)
5	Breastfeeding promotion	+102% EBF at 3 mo (32164187)	High (26619338)
6	SQ-LNS (6–24 mo)	mortality RR 0.73 (31697329)	High (~\$52/child)

What's new in v3 & what's next

- ▶ **Cost-effectiveness is earned, not assumed** — rated only where a real CEA exists; WASH & cash transfers honestly read *Unknown*
- ▶ **Verified & reproducible** — 0 not-in-corpus claims; Cochrane versions counted once; population-targeted throughout
- ▶ **Next milestone** — a trial-level evidence graph to detect shared trials and DEVTA-class dominant-trial effects

How It Works — Two Phases Around a Human Checkpoint

